Type MS · Merge sort: split, then merge

Answer 1 of 2 · Question 2 part c · 5 marks

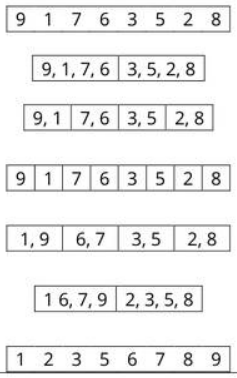
2021 NOV · Paper 02 · extract 1 of 18 · find part c in this mark scheme extract

2 (b)	B1 B2 B3 B4	**Award up to 4 marks for a linked explanation such as:** • The list is not sorted (1) Juan should come after Elija if the list was sorted / Elija is compared with Juan (1), so the bottom half of the list would be discarded (1) after the first pass through the loop (1) • Binary search works on a sorted list (1) because it uses divide and conquer (1) half the list discarded each pass through (1) based on comparison of search item with middle item (1)		(4)
Question	**mp**	**Answer**	**Additional Guidance**	**Mark**

Type MS · Merge sort: split, then merge

Answer 1 of 2 · Question 2 part c · 5 marks

2021 NOV · Paper 02 · extract 2 of 18 · find part c in this mark scheme extract

2 (c)	C1 C2 C3 C4 C5	**Award 1 mark for each of:** Split into two sets of four (1) Split each set into two pairs (1) Split each pair into single elements (1) Merge elements into sorted pairs (1) Merge pairs into sorted sets (1) 	Allow step at C3 to be implied	(5)

Type MS · Merge sort: split, then merge

Answer 1 of 2 · Question 2 part c · 5 marks

2021 NOV · Paper 02 · extract 3 of 18 · find part c in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Merge sort: split, then merge

Answer 1 of 2 · Question 2 part c · 5 marks

2021 NOV · Paper 02 · extract 4 of 18 · find part c in this mark scheme extract

2 (a)	**Award one mark for each of:**		Logic of algorithm must be followed as set out. Alternatives must address each point. Do not penalise candidates who attempt more than the stated requirements. Do not penalise spelling mistakes in the input message.	
	A1	Input as an integer (1)		
	A2	Loop with correct conditions start <= end and not found (1)		
	A3	Use of correct symbol for integer division (in conversion of DIV) or type coercion (1)		
	A4	Check to see if number in middle matches item (1)		
	A5	Set found to true (1)		
	A6	Check to see if item is less than number in middle (1)		
	A7	Set end to middle – 1 (1)		
	A8	else – set start to middle + 1 (1)		
	A9	count incremented by 1 (1)		
	A10	Program execution is fully correct (1)		(10)
Code examples				

Type MS · Merge sort: split, then merge

Answer 1 of 2 · Question 2 part c · 5 marks

2021 NOV · Paper 02 · extract 5 of 18 · find part c in this mark scheme extract

C#

```
Console.WriteLine("What is the number to find? ");
int item = Convert.ToInt32(Console.ReadLine());

while (start <= end && !found)
{
    middle = (start + end) / 2;
    if (numberList[middle] == item)
    {
        found = true;
    }
    else
    {
        if (item < numberList[middle])
        {
            end = middle - 1;
        }
        else
        {
            start = middle + 1;
        }
    }
    count++;
}
```

Type MS · Merge sort: split, then merge

Answer 1 of 2 · Question 2 part c · 5 marks

2021 NOV · Paper 02 · extract 6 of 18 · find part c in this mark scheme extract

Java

```
Scanner input = new Scanner(System.in);
System.out.print("What is the number to find? ");
int item = Integer.parseInt(input.nextLine());

while (start <= end && !found)
{
    middle = (start + end) / 2;
    if (numberList[middle] == item)
    {
        found = true;
    }
    else
    {
      if (item < numberList[middle])
      {
          end = middle - 1;
      }
      else
      {
          start = middle + 1;
      }
    }
    count ++;
}
```

Python

```
item = int(input("What is the number to find? "))
while start <= end and not found:
    middle = (start + end) // 2
    if numberList[middle] == item:
        found = True
    else:
        if item < numberList[middle]:
            end = middle - 1
        else:
            start = middle + 1
    count = count + 1
```

Type MS · Merge sort: split, then merge

Answer 1 of 2 · Question 2 part c · 5 marks

2021 NOV · Paper 02 · extract 7 of 18 · find part c in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Merge sort: split, then merge

Answer 1 of 2 · Question 2 part c · 5 marks

2021 NOV · Paper 02 · extract 8 of 18 · find part c in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
4 (a)	**Award one mark for each of:**			
		Validation		
	A1	Suitable input prompt and storing it in binaryPattern (1)		
	A2	Length of binary pattern validated as 8 (1)		
	A3	Loop used to check the length of binary pattern (1)		
		Binary to denary		
	A4	Loop through each digit in the binary pattern (1)		
	A5	Correct conversion of at least one binary digit to relevant placeholder (1)		
	A6	Attempt at running total using denaryNumber (1)		
	A7	Correct running total using denaryNumber (1)		
	A8	Print statement includes the binary pattern and the denary number (1)		
	A9	Print statement outside of conversion loop (1)		
	A10	Comment explains how conversion works (1)		
	A11	Program executes correctly for any binary pattern that is exactly 8 characters long only (1)		(11)
Code examples				

Type MS · Merge sort: split, then merge

Answer 1 of 2 · Question 2 part c · 5 marks

2021 NOV · Paper 02 · extract 9 of 18 · find part c in this mark scheme extract

C#

```
while (binaryPattern.Length != 8)
{
    Console.WriteLine("Enter an 8 digit binary number to convert to denary ");
    binaryPattern = Console.ReadLine();
}

foreach (char digit in binaryPattern)
{
    if (digit == '1')
    {
        denaryNumber += denaryPlaceholders[count] * 1;
    }
    count++;
}
Console.WriteLine(binaryPattern + " converted to denary is " + denaryNumber);
Console.ReadLine();
```

Java

```
while (binaryPattern.length() != 8)
{
    Scanner input = new Scanner(System.in);
    System.out.print("Enter an 8 digit binary number to convert to denary ");
    binaryPattern = input.next();
}

for (char digit : binaryPattern.toCharArray())
{
    if (digit == '1')
    {
       denaryNumber += denaryPlaceholders[count] * 1;
    }
    count ++;
}
System.out.print(binaryPattern + " converted to denary is " + denaryNumber);
```

Type MS · Merge sort: split, then merge

Answer 1 of 2 · Question 2 part c · 5 marks

2021 NOV · Paper 02 · extract 10 of 18 · find part c in this mark scheme extract

Python

```
while len(binaryPattern) != 8:
    binaryPattern = input("Enter an 8 digit binary number to convert to denary: ")

for digit in binaryPattern:
    if digit == "1":
        denaryNumber = denaryNumber + denaryPlaceholders[count] * 1
    count = count + 1

print(binaryPattern,"converted to denary is", denaryNumber)
```

Type MS · Merge sort: split, then merge

Answer 1 of 2 · Question 2 part c · 5 marks

2021 NOV · Paper 02 · extract 11 of 18 · find part c in this mark scheme extract

For Q5, the first 11 marks are for coding that matches requirements of task. The remaining 9 marks should be allocated on a best fit

Question	mp	Answer	Additional Guidance	Mark
5	A1	At least three variables initialised appropriately		(1)
	A2	Word input		(1)
	A3	Program repeats until 1 is input		(1)
	A4	Loop through each word in the array		(1)
		Words that begin with the same letter		
	A5	Check to see if the first letter of the input word matches the first letter of at least one word in the array		(1)
	A6	The number of words that begin with the same letter displayed		(1)
		Words that contain the input word		
	A7	Check to see if the word contains the input word		(1)
	A8	Number of letters in the word calculated		(1)
	A9	At least one word from the array that contains the input word displayed		(1)
	A10	The number of words that contain the input word displayed		(1)
	A11	Identification of the longest word or the shortest word		(1)

Band 1 (1-3 marks)	Band 2 (4-6 marks)	Band 3 (7-9 marks)	Mark
Little attempt to decompose into component parts	Some attempt to decompose into component parts	The problem has been decomposed into component parts	
Some parts of the logic are clear and appropriate to the problem	Most parts of the logic are clear and mostly appropriate to the problem	The logic is clear and appropriate to the problem	
Some appropriate use and manipulation of data types, variables, data structures and program constructs	The use and manipulation of data types, variables and data structures and program constructs is mostly appropriate	The use and manipulation of data types, variables and data structures and program constructs is appropriate	
Parts of the code are clear and readable	Code is mostly clear and readable	Code is clear and readable	
Finished program will not be flexible enough with other data sets or input	Finished program will function with some but not all other data sets or input	Finished program could be used with other data sets or input	
The program meets some of the given requirements	The program meets most of the given requirements	The program fully meets the given requirements	(9)

Type MS · Merge sort: split, then merge

Answer 1 of 2 · Question 2 part c · 5 marks

2021 NOV · Paper 02 · extract 12 of 18 · find part c in this mark scheme extract

Code examples

C#

```
// Add your code here
string inputWord = "";

while (inputWord != "1")
{
    int shortest = 5000;
    int longest = 0;
    String shortestWord = "";
    String longestWord = "";

    Console.WriteLine("Enter a word or 1 to exit ");
    inputWord = Console.ReadLine();

    if (inputWord != "1")
    {
        int count = 0;

        foreach (String word in wordArray)
        {
            if(word[0] == inputWord[0])
            {
                Console.WriteLine(word);
                count ++;
            }
        } // End of Loop

        Console.WriteLine(count + " word(s) beginning with "+ inputWord[0]);
        Console.WriteLine("------------------------------------------------------------------------");

        count = 0;

        foreach (String word in wordArray)
        {
            if(word.Contains(inputWord))
            {
                count ++;
                Console.WriteLine(word);

                int length = word.Length;

                if (longest < length)
```

Type MS · Merge sort: split, then merge

Answer 1 of 2 · Question 2 part c · 5 marks

2021 NOV · Paper 02 · extract 13 of 18 · find part c in this mark scheme extract

```
                    {
                        longest = length;
                        longestWord = word;
                    }

                    if (shortest > length)
                    {
                        shortest = length;
                        shortestWord = word;
                    }
                }// end of checking word
            } // End of for loop

            // Print the number of words, the number of characters in the longest and
            // shortest word, the longest and the shortest word

            if (count > 0)
            {
                Console.WriteLine(count + " word(s) with "+ inputWord + " in them");
                Console.WriteLine("The longest word has " + longest + " characters");
                Console.WriteLine("The shortest word has " + shortest + " characters");
                Console.WriteLine("The longest word is " + longestWord + " characters");
                Console.WriteLine("The shortest word is " + shortestWord + " characters");
                Console.WriteLine("--------------------------------------------------------------------------------");
            }
            else
            {
                Console.WriteLine("There were 0 words that had all the letters from " + inputWord + " in them");
                Console.WriteLine("--------------------------------------------------------------------------------");
            } // End of printing words
        } // End of check
    } // End of while loop
```

Type MS · Merge sort: split, then merge

Answer 1 of 2 · Question 2 part c · 5 marks

2021 NOV · Paper 02 · extract 14 of 18 · find part c in this mark scheme extract

Java

```
// Add your code here
Scanner input = new Scanner(System.in);
String inputWord = "";

// While there is a word run the program
while (!inputWord.equals("1")) {

    int shortest = 5000;
    int longest = 0;
    String shortestWord = "";
    String longestWord = "";

    System.out.println("Enter a word or 1 to exit ");
    System.out.println("--------------------------------------------------------------------------------");
    inputWord = input.nextLine();
    if (!inputWord.equals("1")){
        // Find words that begin with the same letter as the input word
        int count = 0;

        // Get each word in the array
        for (String word: wordArray){

            // If the first character of the word is the same as the first character of
            // the input word then print the word
            if (word.charAt(0) == inputWord.charAt(0)){
                System.out.println(word);
                count ++;
            }
        } // End of for loop

        // Print the total number of words that begin with the same letter
        System.out.println(count + " word(s) beginning with "+ inputWord.charAt(0));
        System.out.println("--------------------------------------------------------------------------------");

        // Find words that contain the input word
        count = 0;

        // Get each word in the array
        for (String word: wordArray){
            // Check to see if the word contains the inputWord
            if (word.contains(inputWord)){
                count ++;
                System.out.println(word);

                // Set the length of the word
                int length = word.length();
```

Type MS · Merge sort: split, then merge

Answer 1 of 2 · Question 2 part c · 5 marks

2021 NOV · Paper 02 · extract 15 of 18 · find part c in this mark scheme extract



```
                    // If the length is longer than the current longest replace it and the longestWord
                    if (longest < length) {
                        longest = length;
                        longestWord = word;
                    }

                    // If the length is shorter than the shortest replace it and the shortestWord
                    if (shortest > length){
                        shortest = length;
                        shortestWord = word;
                    }
                } // End of checking word
            } // End of for loop

            // Print the number of words, the number of characters in the longest and
            // shortest word, the longest and the shortest word
            if( count > 0){
                System.out.println(count + " word(s) with "+ inputWord + " in them");
                System.out.println("The longest word has " + longest + " characters");
                System.out.println("The shortest word has " + shortest + " characters");
                System.out.println("The longest word is " + longestWord + " characters");
                System.out.println("The shortest word is " + shortestWord + " characters");
                System.out.println("--------------------------------------------------------------------------------");
            } else {
                System.out.println("There were 0 words that had all the letters from " + inputWord + " in them");
                System.out.println("--------------------------------------------------------------------------------");
            } // End of printing words
        }
    } // End of while loop
```

Type MS · Merge sort: split, then merge

Answer 1 of 2 · Question 2 part c · 5 marks

2021 NOV · Paper 02 · extract 16 of 18 · find part c in this mark scheme extract

Python

```
# While there is a word run the program
while inputWord !="1":

    shortest = 5000
    longest = 0
    shortestWord = ""
    longestWord = ""

    inputWord = input("Enter a word or 1 to exit: ")
    print("------------------------------------------------------------------------")

    if inputWord !="1":
        # Find words that begin with the same letter as the input word
        count = 0
        # Get each word in the array
        for word in wordArray:
            # If the first character of the word is the same as the first character of
            # the input word then print the word
            if word[0] == inputWord[0]:
                print(word)
                count = count + 1

        # Print the total number of words that begin with the same letter
        print("\n",count,"word(s) beginning with",inputWord[0])
        print("------------------------------------------------------------------------")

        # Find words that contain the input word
        count = 0
        # Get each word in the array
        for word in wordArray:
            # Check to see if the word contains the inputWord
            if inputWord in word:
                count = count + 1
                print(word)
                # Set the length of the word
                length = len(word)

                # If the length is longer than the current longest replace it and the longestWord
                if longest < length:
                    longest = length
                    longestWord = word

                # If the length is shorter than the shortest replace it and the shortestWord
                if shortest > length:
                    shortest = length
                    shortestWord = word
```

Type MS · Merge sort: split, then merge

Answer 1 of 2 · Question 2 part c · 5 marks

2021 NOV · Paper 02 · extract 17 of 18 · find part c in this mark scheme extract

```
59
60          # Print the number of words, the number of characters in the longest and
61          # shortest word, the longest and the shortest word
62          if count > 0:
63              print("\n",count,"word(s) with",inputWord,"in them")
64              print("The longest word has",longest,"letters")
65              print("The shortest word has",shortest,"letters")
66              print("The longest word is",longestWord)
67              print("The shortest word is",shortestWord)
68              print("-------------------------------------------------------------------------")
69          else:
70              print("There were 0 words with all the letters from",inputWord,"in them")
71          # ------------------------------------------------------------------------------------
72
73  # End the program when there are no more words
74  print("End of program")
```

Type MS · Merge sort: split, then merge

Answer 1 of 2 · Question 2 part c · 5 marks

2021 NOV · Paper 02 · extract 18 of 18 · find part c in this mark scheme extract

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Type MS · Merge sort: split, then merge

Answer 2 of 2 · Question 5 part a · 2 marks

2023 JUN · Paper 02 · extract 1 of 9 · find part a in this mark scheme extract

5(a)	A1 A2	Award 1 mark for demonstration of one correctly merged/split row Award 2 marks for correct merge sort Blackfin, Bigeye | Longtail, Albacore | Bluefin Longtail, Blackfin, Bigeye, Albacore | Bluefin Longtail, Bluefin, Blackfin, Bigeye, Albacore OR Bigeye | Blackfin, Albacore | Longtail, Bluefin Bigeye | Longtail, Bluefin, Blackfin, Albacore Longtail, Bluefin, Blackfin, Bigeye, Albacore OR Blackfin, Bigeye | Albacore | Longtail, Bluefin Blackfin, Bigeye | Longtail, Bluefin, Albacore	We are testing pupils understanding of how a merge sort progresses: • Divide and conquer (split) • Merging adjacent elements Allow answers in ascending order	**2**

Type MS · Merge sort: split, then merge

Answer 2 of 2 · Question 5 part a · 2 marks

2023 JUN · Paper 02 · extract 2 of 9 · find part a in this mark scheme extract

		Longtail, Bluefin, Blackfin, Bigeye, Albacore OR Blackfin, Bigeye | Albacore | Longtail, Bluefin Blackfin, Bigeye, Albacore | Longtail, Bluefin Longtail, Bluefin, Blackfin, Bigeye, Albacore		

Type MS · Merge sort: split, then merge

Answer 2 of 2 · Question 5 part a · 2 marks

2023 JUN · Paper 02 · extract 3 of 9 · find part a in this mark scheme extract

<table>
<tr><th>Question</th><th>mp</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>5(b)</td><td>B1
B2</td><td>Award one mark for each correct cell:
<table><tr><td>Line number with error</td><td>11 (1)</td></tr><tr><td>Corrected line of pseudocode</td><td>SET tmp TO myTuna[ndx] (1)</td></tr></table></td><td>Ignore spellings

Ignore missing text as long as the minimum of [ndx] is provided

B2 does not depend on B1</td><td>2</td></tr>
</table>

Question	mp	Answer	Additional guidance	Mark
5(c)	Award **one** mark for each of:		C1 allow append as writing to file C4 Do not award comma within an array C5 must have 1 complete line written to file (allow additional spaces and commas). Should include individual elements (2D index) and not a 1D array from tbl_tuna C6 complete file must be as expected e.g. no additional comma at end of line and no spaces around commas	**6**

Type MS · Merge sort: split, then merge

Answer 2 of 2 · Question 5 part a · 2 marks

2023 JUN · Paper 02 · extract 4 of 9 · find part a in this mark scheme extract

C#

```
// Q05cFINISHED

using System.IO;

String[,] tblTuna =  {
    { "Yellowfin", "105", "15", "3"},
    { "Albacore","90","15","5"},
    { "Skipjack","50","3","4"},
    { "Bigeye","105","25","4"},
    { "Atlantic Bonito","50","4","2"},
    { "Northern Bluefin","190","120","11"},
    { "Southern Bluefin","190","120","11"},
    { "Tongol","90","20","4"}
};



// ------------------------------------------
// Write your code below this line

String fileName = "TunaData.txt";       // Output file name
String comma = ",";                     // Use as a constant
String full_path = "C:\\Q05c";
fileName = full_path + "\\" + fileName;

// Create variables as needed
int number = 101;           // Number for the line at the front
String lineOut = "";        // The line to output

// Get the dimensions of the array for looping
int rows = tblTuna.GetLength(0);
int columns = tblTuna.GetLength(1);

// Open the file for writing
StreamWriter fileWriter = new StreamWriter(fileName);

```

Type MS · Merge sort: split, then merge

Answer 2 of 2 · Question 5 part a · 2 marks

2023 JUN · Paper 02 · extract 5 of 9 · find part a in this mark scheme extract

```
// Process every tuna in the table
for (int i = 0; i < rows; i++)          // Rows
{
    // Start with the number and a comma
    lineOut = number.ToString() + comma;

    for (int j = 0; j < columns; j++)   // Columns
    {
        // Add the name and the numbers
        lineOut = lineOut + tblTuna[i, j];

        // Add a comma to all the fields except the last
        if (j < columns - 1)
        {
            lineOut = lineOut + comma;
        }
    }
    // Write the line to the file
    fileWriter.WriteLine(lineOut);

    // Go to the next number for the line
    number = number + 1;
}

// Close the file
fileWriter.Close();
```

Type MS · Merge sort: split, then merge

Answer 2 of 2 · Question 5 part a · 2 marks

2023 JUN · Paper 02 · extract 6 of 9 · find part a in this mark scheme extract

Java

```java
// Q05cFINISHED

import java.io.FileWriter;
import java.io.IOException;

public class Q05cFINISHED {

    public static void main(String[] args) throws Exception {

        String[][] tblTuna =
        {
            { "Yellowfin", "105", "15", "3"},
            { "Albacore","90","15","5"},
            { "Skipjack","50","3","4"},
            { "Bigeye","105","25","4"},
            { "Atlantic Bonito","50","4","2"},
            { "Northern Bluefin","190","120","11"},
            { "Southern Bluefin","190","120","11"},
            { "Tongol","90","20","4"}
        };

        // ------------------------------------------
        // Write your code below this
        String fileName = "TunaData.txt";       // Output file name
        String comma = ",";                     // Use as a constant
        String full_path = "C:\\src\\q05c";
        fileName = full_path + "\\" + fileName;

        // Create variables as needed
        int number = 101;           // Number for the line at the front
        String lineOut = "";        // The line to output
        String linefeed = "\n";     // Use as a constant

        // Get the dimensions of the array for looping
        int rows = tblTuna.length;
        int columns = tblTuna[0].length;

        // Open the file for writing
        FileWriter fileWriter = new FileWriter(fileName);

```

Type MS · Merge sort: split, then merge

Answer 2 of 2 · Question 5 part a · 2 marks

2023 JUN · Paper 02 · extract 7 of 9 · find part a in this mark scheme extract

```
41          // Process every tuna in the table
42          for (int i = 0; i < rows; i++)          // Rows
43          {
44              // Start with the number and a comma
45              lineOut = Integer.toString(number) + comma;
46
47              for (int j = 0; j < columns; j++)   // Columns
48              {
49                  // Add the name and the numbers
50                  lineOut = lineOut + tblTuna[i][j];
51
52                  // Add a comma to all the fields except the last
53                  if (j < columns - 1)
54                  {
55                      lineOut = lineOut + comma;
56                  }
57                  else
58                  {
59                      lineOut = lineOut + linefeed;
60                  }
61              }
62              // Write the line to the file
63              fileWriter.write(lineOut);
64
65              // Go to the next number for the line
66              number = number + 1;
67          }
68
69          // Close the file
70          fileWriter.close();
71      }
72
73  }
```

Type MS · Merge sort: split, then merge

Answer 2 of 2 · Question 5 part a · 2 marks

2023 JUN · Paper 02 · extract 8 of 9 · find part a in this mark scheme extract

Python

```
# Q05cFINISHED

tbl_tuna = [["Yellowfin",105,15,3],
            ["Albacore",90,15,5],
            ["Skipjack",50,3,4],
            ["Bigeye",105,25,4],
            ["Atlantic Bonito",50,4,2],
            ["Northern Bluefin",190,120,11],
            ["Southern Bluefin",190,120,11],
            ["Tongol",90,20,4]]

# ---------------------------------------------
# Write your code below this line

FILE_OUT = "TunaData.txt"           # Output file name
COMMA = ","                         # Use as constant

# Create variables as needed
LINEFEED = "\n"
number = 101                        # Number at the front
line_out = ""                       # The line to write

# Open the file for writing
file = open (FILE_OUT, "w")

# Process every tuna in the table
for tuna in tbl_tuna:
    # Start with the number and a comma
    line_out = str (number) + COMMA

    # Add the name of the tuna and a comma
    line_out = line_out + tuna[0] + COMMA

    # Each of the numbers need to be converted to a string
    # before adding to the output string.
    line_out = line_out + str (tuna[1]) + COMMA
    line_out = line_out + str (tuna[2]) + COMMA
    line_out = line_out + str (tuna[3])     # No comma on last field

    # Add a line feed to the whole line
    line_out = line_out + LINEFEED

    # Write the line to the file
    file.write (line_out)

    # Go to the next number for the line
    number = number + 1

# Close the file
file.close ()
```

Type MS · Merge sort: split, then merge

Answer 2 of 2 · Question 5 part a · 2 marks

2023 JUN · Paper 02 · extract 9 of 9 · find part a in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark